1) a)

$$\pi\int_1^{14} y^2\,\mathrm{d}x = \pi\int_1^{14} (2x-1)^{\frac{2}{3}}\,\mathrm{d}x$$

$$= \pi\left[\tfrac{3}{10}(2x-1)^{\frac{5}{3}}\right]_1^{14} = \pi\left[\left(\tfrac{3}{10}\times 243\right) - \left(\tfrac{3}{10}\right)\right] = \tfrac{363}{5}\pi \text{ units}^3$$

1) b)

$$\pi\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} y^2\,\mathrm{d}x = \pi\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x\,\mathrm{d}x$$

$$= \pi[\tan x]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \pi\left[\tan\tfrac{\pi}{3} - \tan\tfrac{\pi}{6}\right] = \pi\left(\sqrt{3} - \tfrac{\sqrt{3}}{3}\right) = \tfrac{2\sqrt{3}}{3}\pi \text{ units}^3$$

1) c)

$$\pi\int_1^2 y^2\,\mathrm{d}x = \pi\int_1^2 \ln x\,\mathrm{d}x$$

Use Integration by Parts:

$$u = \ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad v = x$$

$$\therefore \int 1\times\ln x\,\mathrm{d}x = x\ln x - \int 1\,\mathrm{d}x$$

$$= x\ln x - x + c$$

So that:

$$\pi\int_1^2 \ln x\,\mathrm{d}x = \pi[x\ln x - x]_1^2 = \pi[(2\ln 2 - 2) - (0-1)] = \pi(\ln 4 - 1) \text{ units}^3$$

1) d)

$$\pi\int_0^{\frac{\pi}{2}} y^2\,\mathrm{d}x = \pi\int_0^{\frac{\pi}{2}} \frac{2\cos x}{1+\sin x}\,\mathrm{d}x$$

$$= \pi[2\ln(1+\sin x)]_0^{\frac{\pi}{2}} = \pi[(2\ln 2) - (0)] = \pi\ln 4 \text{ units}^3$$

1) e)

$$\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} y^2\,\mathrm{d}x = \pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x\,\mathrm{cosec}^2 x\,\mathrm{d}x$$

Use Integration by Parts:

$$u = x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{cosec}^2 x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 1 \qquad v = -\cot x$$

$$\therefore \int x\,\mathrm{cosec}^2 x\,\mathrm{d}x = -x\cot x + \int \cot x\,\mathrm{d}x = -x\cot x + \int \frac{\cos x}{\sin x}\,\mathrm{d}x$$

$$= -x\cot x + \ln|\sin x| + c$$

So that:

$$\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} x\,\mathrm{cosec}^2 x\,\mathrm{d}x = \pi[-x\cot x + \ln|\sin x|]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \pi\left[(0+0) - \left(-\frac{\pi}{4} - \ln\frac{\sqrt{2}}{2}\right)\right] = \left(\frac{\pi}{4} + \frac{1}{2}\ln 2\right)\pi \text{ units}^3$$

1) f)

$$\pi\int_0^{\frac{\pi}{4}} y^2\,\mathrm{d}x = \pi\int_0^{\frac{\pi}{4}} (\sec x + \tan x)^2\,\mathrm{d}x = \pi\int_0^{\frac{\pi}{4}} (\sec^2 x + 2\sec x\tan x + \tan^2 x)\,\mathrm{d}x$$

$$= \pi\int_0^{\frac{\pi}{4}} (\sec^2 x + 2\sec x\tan x + \sec^2 x - 1)\,\mathrm{d}x = \pi\int_0^{\frac{\pi}{4}} (2\sec^2 x + 2\sec x\tan x - 1)\,\mathrm{d}x$$

$$= \pi[2\tan x + 2\sec x - x]_0^{\frac{\pi}{4}}$$

$$= \pi\left[\left(2 + \frac{4}{\sqrt{2}} - \frac{\pi}{4}\right) - (0 + 2 - 0)\right] = \left(2\sqrt{2} - \frac{\pi}{4}\right)\pi \text{ units}^3$$

2) a)

When $y = 0$ then $\sqrt{\cos x} = 0$ so $x = \frac{\pi}{2}$

$$\pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} y^2\,\mathrm{d}x = \pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\sqrt{\cos x}\right)^2\,\mathrm{d}x = \pi\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x\,\mathrm{d}x$$

$$= \pi[\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \pi\left[\sin\frac{\pi}{2} - \sin\frac{\pi}{4}\right] = \left(1 - \frac{\sqrt{2}}{2}\right)\pi \text{ units}^3$$

2) b)

When $y = 0$ then $\sqrt{x}\ln x = 0$ so $x = 0$ or $x = 1$

$$\pi\int_1^4 y^2\,\mathrm{d}x = \pi\int_1^4 x(\ln x)^2\,\mathrm{d}x$$

Use Integration by Parts:

$$u = (\ln x)^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2\ln x}{x} \qquad v = \frac{1}{2}x^2$$

$$\therefore \int x(\ln x)^2\,\mathrm{d}x = \tfrac{1}{2}x^2(\ln x)^2 - \int x\ln x\,\mathrm{d}x$$

Use Integration by Parts (again):

$$u = \ln x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \qquad v = \frac{1}{2}x^2$$

$$\therefore \int x\ln x\,\mathrm{d}x = \tfrac{1}{2}x^2\ln x - \tfrac{1}{2}\int x\,\mathrm{d}x = \tfrac{1}{2}x^2\ln x - \tfrac{1}{4}x^2(+c)$$

$$\therefore \int x(\ln x)^2\,\mathrm{d}x = \tfrac{1}{2}x^2(\ln x)^2 - \left(\tfrac{1}{2}x^2\ln x - \tfrac{1}{4}x^2\right) = \tfrac{1}{2}x^2(\ln x)^2 - \tfrac{1}{2}x^2\ln x + \tfrac{1}{4}x^2$$

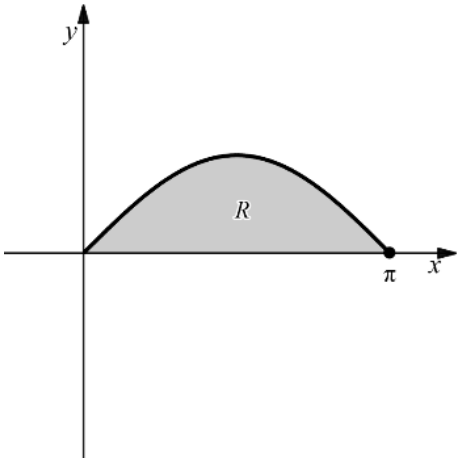
So that:

$$\pi\int_1^4 x(\ln x)^2\,\mathrm{d}x$$

$$= \pi\left[\frac{1}{2}x^2(\ln x)^2 - \frac{1}{2}x^2\ln x + \frac{1}{4}x^2\right]_1^4$$

$$= \pi\left[(8(\ln 4)^2 - 8\ln 4 + 4) - \left(0 - 0 + \tfrac{1}{4}\right)\right] = \left(8(\ln 4)^2 - 8\ln 4 + \tfrac{15}{4}\right)\pi \text{ units}^3$$

3) a)



3) b)

$$\pi\int_0^{\pi} y^2\,\mathrm{d}x = \pi\int_0^{\pi} (\sin x)^2\,\mathrm{d}x$$

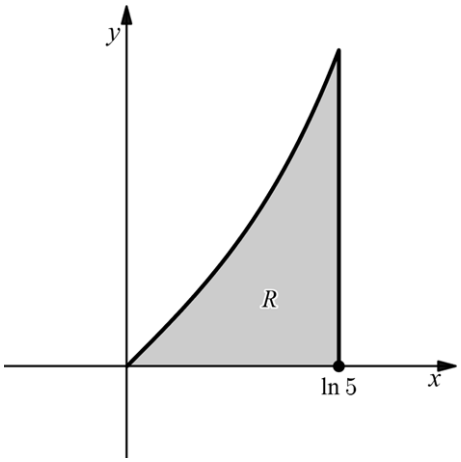
Since $\cos 2A = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$ so that

$$\pi\int_0^{\pi} (\sin x)^2\,\mathrm{d}x = \pi\int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)\mathrm{d}x$$

$$= \pi\left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_0^{\pi}$$

$$= \pi\left[\left(\frac{\pi}{2} - 0\right) - (0 - 0)\right] = \frac{1}{2}\pi^2 \text{ units}^3$$

4) a)



4) b)

$$\pi\int_0^{\pi} y^2\,\mathrm{d}x = \pi\int_0^{\ln 5} (\sinh x)^2\,\mathrm{d}x$$

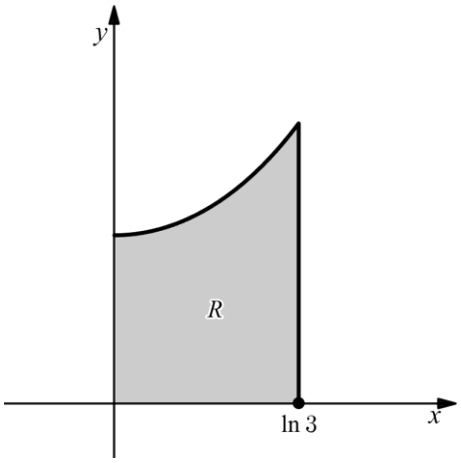
Since $\cosh 2A = 1 + 2\sinh^2 A$ then $\sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$ so that

$$\pi\int_0^{\ln 5} (\sinh x)^2\,\mathrm{d}x = \pi\int_0^{\ln 5} \left(\frac{1}{2}\cosh 2x - \frac{1}{2}\right)\mathrm{d}x$$

$$= \pi\left[\frac{1}{4}\sinh 2x - \frac{1}{2}x\right]_0^{\ln 5}$$

$$= \pi\left[\left(\frac{1}{4}\sinh(2\ln 5) - \frac{1}{2}\ln 5\right) - (0 - 0)\right] = \left(\frac{78}{25} - \frac{1}{2}\ln 5\right)\pi \text{ units}^3$$

5) a)



5) b)

$\pi \int_0^{\pi} y^2 \, dx = \pi \int_0^{\ln 3} (\cosh x)^2 \, dx$

Since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$ so that

$\pi \int_0^{\ln 3} (\cosh x)^2 \, dx = \pi \int_0^{\ln 3} \left(\frac{1}{2}\cosh 2x + \frac{1}{2}\right) dx$

$= \pi \left[\frac{1}{4}\sinh 2x + \frac{1}{2}x\right]_0^{\ln 3}$

$= \pi \left[\left(\frac{1}{4}\sinh(2\ln 3) + \frac{1}{2}\ln 3\right) - (0 + 0)\right] = \left(\frac{10}{9} + \frac{1}{2}\ln 3\right)\pi$ units^3

6) a)

$2x\cos x = 0$ so $x = \frac{\pi}{2} \therefore P = \frac{\pi}{2}$

6) b)

$\pi \int_0^{\frac{\pi}{2}} 2x \cos x \, \mathrm{d}x$

Use Integration by Parts:

$u = 2x$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 2$ $\qquad$ $v = \sin x$

$\therefore \int 2x \cos x \, \mathrm{d}x = 2x \sin x - 2 \int \sin x \, \mathrm{d}x = 2x \sin x + 2 \cos x \, (+c)$

So that:

$\pi \int_0^{\frac{\pi}{2}} 2x \cos x \, \mathrm{d}x = \pi [2x \sin x + 2 \cos x]_0^{\frac{\pi}{2}}$

$= \pi[(\pi + 0) - (0 + 2)] = (\pi^2 - 2\pi)$ units^3

7) a)

$a = 1$

7) b)

$2.5 = 1 + k(6)^2 \Rightarrow k = \frac{2.5-1}{36} = \frac{1}{24}$ as req.

$2.5 = 1 + m\left(1 - \cos\left(\frac{6\pi}{6}\right)\right) \Rightarrow m = \frac{2.5-1}{1-\cos\pi} = \frac{3}{4}$ as req.

7) c)

Model A:

$$\pi\int_{-6}^{6} y^2\,\mathrm{d}x = \pi\int_{-6}^{6}\left(1+\frac{1}{24}x^2\right)^2\mathrm{d}x$$

$$= \pi\int_{-6}^{6}\left(1+\frac{1}{12}x^2+\frac{1}{576}x^4\right)\mathrm{d}x$$

$$= \pi\left[x+\frac{1}{36}x^3+\frac{1}{2880}x^5\right]_{-6}^{6}$$

$$= \pi\left[\left(6+6+\tfrac{27}{10}\right)-\left(-6-6-\tfrac{27}{10}\right)\right]$$

$$= \frac{147}{5}\pi$$

$$= 92.4\ \text{cm}^3$$

Model B:

$$\pi\int_{-6}^{6} y^2\,\mathrm{d}x = \pi\int_{-6}^{6}\left(1+\frac{3}{4}\left(1-\cos\frac{\pi x}{6}\right)\right)^2\mathrm{d}x = \pi\int_{-6}^{6}\left(\frac{7}{4}-\frac{3}{4}\cos\frac{\pi x}{6}\right)^2\mathrm{d}x$$

$$= \pi\int_{-6}^{6}\left(\frac{49}{16}-\frac{21}{8}\cos\left(\frac{\pi x}{6}\right)+\frac{9}{16}\cos^2\left(\frac{\pi x}{6}\right)\right)\mathrm{d}x$$

Since $\cos 2A = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2}+\frac{1}{2}\cos 2A$

$$\therefore \pi\int_{-6}^{6}\left(\frac{49}{16}-\frac{21}{8}\cos\left(\frac{\pi x}{6}\right)+\frac{9}{16}\cos^2\left(\frac{\pi x}{6}\right)\right)\mathrm{d}x$$

$$= \pi\int_{-6}^{6}\left(\frac{49}{16}-\frac{21}{8}\cos\left(\frac{\pi x}{6}\right)+\frac{9}{32}+\frac{9}{32}\cos\left(\frac{\pi x}{3}\right)\right)\mathrm{d}x$$

$$= \pi\int_{-6}^{6}\left(\frac{107}{32}-\frac{21}{8}\cos\left(\frac{\pi x}{6}\right)+\frac{9}{32}\cos\left(\frac{\pi x}{3}\right)\right)\mathrm{d}x$$

$$= \pi\left[\frac{107}{32}x-\frac{126}{8\pi}\sin\left(\frac{\pi x}{6}\right)+\frac{27}{32\pi}\sin\left(\frac{\pi x}{3}\right)\right]_{-6}^{6}$$

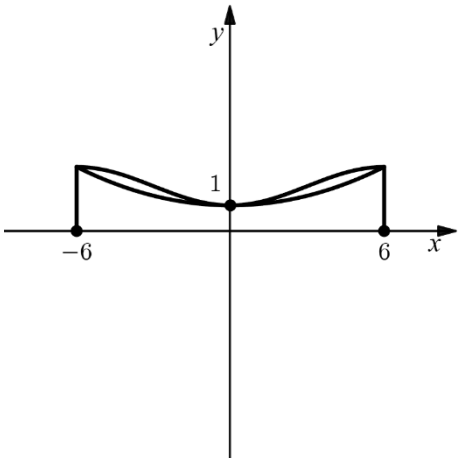
$$= \pi\left[\tfrac{321}{16}-\left(-\tfrac{321}{16}\right)\right]$$

$$= \frac{321}{8}\pi$$

$$= 126\ \text{cm}^3$$

Model A is 75.4 cm^{3}and Model B is 126 cm^3

7) d)



Model A is a quadratic: $y = 1 + \frac{1}{24}x^2$, $\frac{dy}{dx} = \frac{1}{12}x$ so the rate at which the radius changes with x increases linearly, so it increases continuously from the midpoint to the ends, so the curve becomes steeper towards the ends.
In Model B the gradient is zero at both the midpoint and the ends, so the curve flattens again near the ends, which does not match the requirement.
∴ Model A is more appropriate.

8) a)

$\frac{6x+11}{(x+3)(2x-1)} \equiv \frac{A}{x+3} + \frac{B}{2x-1}$

$6x + 11 = A(2x - 1) + B(x + 3)$

Let $x = -3$: $\quad -7 = -7A \therefore A = 1$

Let $x = \frac{1}{2}$: $\quad 14 = \frac{7}{2}B \therefore B = 4$

$\therefore A = 1$ and $B = 4$

$\frac{6x+11}{(x+3)(2x-1)} = \frac{1}{x+3} + \frac{4}{2x-1}$

8) b)

$$\pi \int_1^4 y^2 \, dx = \pi \int_1^4 \left(\sqrt{\frac{6x+11}{(x+3)(2x-1)}} \right)^2 dx = \pi \int_1^4 \frac{6x+11}{(x+3)(2x-1)} \, dx$$

$$= \pi \int_1^4 \left(\frac{1}{x+3} + \frac{4}{2x-1} \right) dx$$

$$= \pi [\ln|x+3| + 2\ln|2x-1|]_1^4$$

$$= \pi[(\ln 7 + 2\ln 7) - (\ln 4 + 0)] = \left(\ln \frac{343}{4}\right)\pi \text{ units}^3$$

9)

$$\pi\int_1^3 y^2\,\mathrm{d}x = \pi\int_1^3 \left(\frac{3x+4}{(x+2)\sqrt{x}}\right)^2 \mathrm{d}x = \pi\int_1^3 \left(\frac{9x^2+24x+16}{(x+2)^2x}\right)\mathrm{d}x$$

$\frac{9x^2+24x+16}{x(x+2)^2} \equiv \frac{A}{x}+\frac{B}{x+2}+\frac{C}{(x+2)^2}$

$9x^2+24x+16 = A(x+2)^2+Bx(x+2)+Cx$

Let $x=0$: $\quad 16 = 4A \therefore A = 4$

Let $x=-2$: $\quad 4 = -2C \therefore C = -2$

Let $x=1$: $\quad 49 = 3B+34 \therefore B = 5$

$\therefore A = 4$, $B = 5$ and $C = -2$

$\frac{9x^2+24x+16}{x(x+2)^2} = \frac{4}{x}+\frac{5}{x+2}-\frac{2}{(x+2)^2}$

So that

$$\pi\int_1^3 \left(\frac{9x^2+24x+16}{(x+2)^2x}\right)\mathrm{d}x = \pi\int_1^3 \left(\frac{4}{x}+\frac{5}{x+2}-\frac{2}{(x+2)^2}\right)\mathrm{d}x$$

$$= \pi\left[4\ln|x|+5\ln|x+2|+\frac{2}{x+2}\right]_1^3$$

$$= \pi\left[\left(4\ln 3+5\ln 5+\frac{2}{5}\right)-\left(0+5\ln 3+\frac{2}{3}\right)\right]$$

$$= \pi\left[-\ln 3+5\ln 5-\frac{4}{15}\right]$$

$$= \pi\left(\ln\frac{5^5}{3}-\frac{4}{15}\right)\text{ units}^3$$

10) a)

$$\pi\int_0^{\frac{k}{2}} y^2\,\mathrm{d}x = \pi\int_0^{\frac{k}{2}}\left(\frac{1}{\sqrt{k-x^2}}\right)^2\mathrm{d}x = \pi\int_0^{\frac{k}{2}}\frac{1}{k-x^2}\,\mathrm{d}x$$

$$\frac{1}{(\sqrt{k}+x)(\sqrt{k}-x)} \equiv \frac{A}{\sqrt{k}+x}+\frac{B}{\sqrt{k}-x}$$

$$1 = A\left(\sqrt{k}-x\right)+B\left(\sqrt{k}+x\right)$$

Let $x=-\sqrt{k}$: $\quad 1 = 2\sqrt{k}A \therefore A = \frac{1}{2\sqrt{k}}$

Let $x=\sqrt{k}$: $\quad 1 = 2\sqrt{k}B \therefore B = \frac{1}{2\sqrt{k}}$

$$\frac{1}{k-x^2} = \frac{1}{2\sqrt{k}(\sqrt{k}+x)}+\frac{1}{2\sqrt{k}(\sqrt{k}-x)}$$

So that

$$\pi\int_0^{\frac{k}{2}}\frac{1}{k-x^2}\,\mathrm{d}x = \pi\int_0^{\frac{k}{2}}\left(\frac{1}{2\sqrt{k}(\sqrt{k}+x)}+\frac{1}{2\sqrt{k}(\sqrt{k}-x)}\right)\mathrm{d}x$$

$$= \pi\left[\frac{1}{2\sqrt{k}}\ln\left|\sqrt{k}+x\right|-\frac{1}{2\sqrt{k}}\ln\left|\sqrt{k}-x\right|\right]_0^{\frac{k}{2}}$$

$$= \frac{\pi}{2\sqrt{k}}\left[\ln\left|\sqrt{k}+x\right|-\ln\left|\sqrt{k}-x\right|\right]_0^{\frac{k}{2}}$$

$$= \frac{\pi}{2\sqrt{k}}\left[\ln\left|\frac{\sqrt{k}+x}{\sqrt{k}-x}\right|\right]_0^{\frac{k}{2}}$$

Since $\frac{k}{2} < \sqrt{k}$ when $0<k<4$ then $x<\sqrt{k}$ so $\sqrt{k}+x>0$ and $\sqrt{k}-x<0$

$$\therefore \pi\int_0^{\frac{k}{2}}\frac{1}{k-x^2}\,\mathrm{d}x = \frac{\pi}{2\sqrt{k}}\left[\ln\left(\frac{\sqrt{k}+x}{\sqrt{k}-x}\right)\right]_0^{\frac{k}{2}}$$

$$= \frac{\pi}{2\sqrt{k}}\left[\ln\left(\frac{\sqrt{k}+\frac{k}{2}}{\sqrt{k}-\frac{k}{2}}\right)-\ln\left(\frac{\sqrt{k}}{\sqrt{k}}\right)\right]$$

$$= \frac{\pi}{2\sqrt{k}}\left[\ln\left(\frac{\sqrt{k}+\frac{k}{2}}{\sqrt{k}-\frac{k}{2}}\right)\right]$$

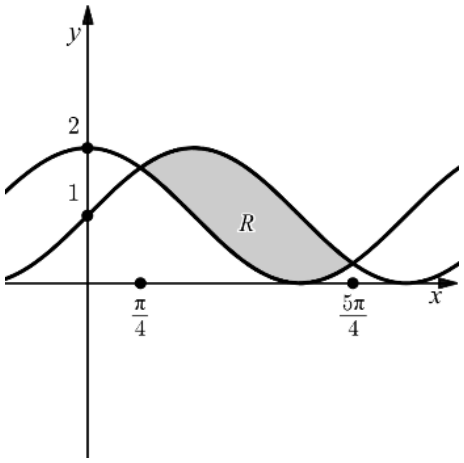
$$= \frac{\pi}{2\sqrt{k}}\left(\ln\left(\frac{\sqrt{k}+\frac{k}{2}}{\sqrt{k}-\frac{k}{2}}\right)\right) \text{ units}^3 \text{ as req.}$$

10) b)

Set $k=1$:

$\frac{\pi}{2}\ln 3$ units^3

11) a)



11) b)

$$\pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} y^2 \, \mathrm{d}x = \pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (1+\sin x)^2 \, \mathrm{d}x - \pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (1+\cos x)^2 \, \mathrm{d}x$$

$$= \pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} ((1+\sin x)^2 - (1+\cos x)^2) \, \mathrm{d}x$$

$$= \pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (1 + 2\sin x + \sin^2 x - 1 - 2\cos x - \cos^2 x) \, \mathrm{d}x$$

Since $\cos^2 x - \sin^2 x = \cos 2x$ this gives:

$$= \pi \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (2\sin x - 2\cos x - \cos 2x) \, \mathrm{d}x$$

$$= \pi \left[-2\cos x - 2\sin x - \frac{1}{2}\sin 2x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= \pi \left[\left(-2\cos\frac{5\pi}{4} - 2\sin\frac{5\pi}{4} - \frac{1}{2}\sin\frac{5\pi}{2} \right) - \left(-2\cos\frac{\pi}{4} - 2\sin\frac{\pi}{4} - \frac{1}{2}\sin\frac{\pi}{2} \right) \right]$$

$$= \pi \left[\left(\sqrt{2} + \sqrt{2} - \tfrac{1}{2} \right) - \left(-\sqrt{2} - \sqrt{2} - \tfrac{1}{2} \right) \right]$$

$$= \pi \left(4\sqrt{2} \right)$$

$$= 4\sqrt{2}\pi$$

12) a)

$\left(z-\frac{1}{z}\right)^4 = z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4} = \left(z^4 + \frac{1}{z^4}\right) - 4\left(z^2 + \frac{1}{z^2}\right) + 6$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z-\frac{1}{z}\right)^4 = 2\cos 4\theta - 4 \times 2\cos 2\theta + 6$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

So $(2\mathrm{i}\sin\theta)^4 = 2\cos 4\theta - 8\cos 2\theta + 6$

$16\sin^4\theta = 2\cos 4\theta - 8\cos 2\theta + 6$

$\therefore \sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}$

$\therefore a = \frac{1}{8}$, $b = -\frac{1}{2}$ and $c = \frac{3}{8}$

12) b)

$\pi\int_0^{\pi} y^2\,\mathrm{d}x = \pi\int_0^{\pi}(\sin^2 x + 1)^2\,\mathrm{d}x = \pi\int_0^{\pi}(\sin^4 x + 2\sin^2 x + 1)\,\mathrm{d}x$

Since $\cos 2A = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$

and $\sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}$:

$\pi\int_0^{\pi}(\sin^4 x + 2\sin^2 x + 1)\,\mathrm{d}x$

$= \pi\int_0^{\pi}\left(\frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8} + 1 - \cos 2x + 1\right)\mathrm{d}x$

$= \pi\int_0^{\pi}\left(\frac{1}{8}\cos 4x - \frac{3}{2}\cos 2x + \frac{19}{8}\right)\mathrm{d}x$

$= \pi\left[\frac{1}{32}\sin 4x - \frac{3}{4}\sin 2x + \frac{19}{8}x\right]_0^{\pi}$

$= \pi\left[\left(0 - 0 + \frac{19}{8}\pi\right) - (0 - 0 + 0)\right] = \frac{19}{8}\pi^2$ units^3

13) a)

$u = 2 - \cos x$ so $\frac{\mathrm{d}u}{\mathrm{d}x} = \sin x$

When $x = 0$, $u = 1$ and when $x = \pi$, $u = 3$

$\int_0^{\pi}\frac{\sin x}{\sqrt{2-\cos x}}\mathrm{d}x = \int_1^3\frac{1}{\sqrt{u}}\mathrm{d}u = \int_1^3 u^{-\frac{1}{2}}\mathrm{d}u$

$= \left[2u^{\frac{1}{2}}\right]_1^3 = 2\sqrt{3} - 2$

13) b)

$$\pi\int_0^{\pi} y^2\,\mathrm{d}x = \pi\int_0^{\pi}\left(2(\sin x)^{\frac{1}{2}}(2-\cos x)^{-\frac{1}{4}}\right)^2\mathrm{d}x = 4\pi\int_0^{\pi}\left(\sin x\,(2-\cos x)^{-\frac{1}{2}}\right)\mathrm{d}x$$

$$= 4\pi\int_0^{\pi}\frac{\sin x}{\sqrt{2-\cos x}}\mathrm{d}x = 4\pi\left(2\sqrt{3}-2\right)\text{ units}^3$$

14)

$$\pi\int_0^{\ln k} y^2\,\mathrm{d}x = \pi\int_0^{\ln k}\left(\sqrt{x}\mathrm{e}^x\right)^2\mathrm{d}x = \pi\int_0^{\ln k} x\mathrm{e}^{2x}\,\mathrm{d}x$$

Use Integration by Parts:

$$u = x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 1 \qquad v = \frac{1}{2}\mathrm{e}^{2x}$$

$$\therefore \int x\mathrm{e}^{2x}\,\mathrm{d}x = \tfrac{1}{2}x\mathrm{e}^{2x} - \tfrac{1}{2}\int \mathrm{e}^{2x}\,\mathrm{d}x = \tfrac{1}{2}x\mathrm{e}^{2x} - \tfrac{1}{4}\mathrm{e}^{2x}(+c)$$

So that:

$$\pi\int_0^{\ln k} x\mathrm{e}^{2x}\,\mathrm{d}x = \pi\left[\frac{1}{2}x\mathrm{e}^{2x} - \frac{1}{4}\mathrm{e}^{2x}\right]_0^{\ln k} = \pi\left[\left(\frac{1}{2}\ln k\times\mathrm{e}^{2\ln k} - \frac{1}{4}\mathrm{e}^{2\ln k}\right) - \left(0-\frac{1}{4}\right)\right]$$

$$= \pi\left[\frac{1}{2}\ln k\times\mathrm{e}^{\ln k^2} - \frac{1}{4}\mathrm{e}^{\ln k^2} + \frac{1}{4}\right]$$

$$= \pi\left[\frac{1}{2}k^2\ln k - \frac{1}{4}k^2 + \frac{1}{4}\right]$$

Given that $81\ln 3 - 20 = \frac{1}{2}k^2\ln k - \frac{1}{4}k^2 + \frac{1}{4}$

Then $-20 = -\frac{1}{4}k^2 + \frac{1}{4} \Rightarrow -\frac{81}{4} = -\frac{1}{4}k^2 \Rightarrow 81 = k^2 \Rightarrow$ Since k must be positive $k =$So

$$\frac{1}{2}k^2\ln k = \frac{81}{2}\ln 9 = 81\ln 3$$

$$\therefore k = 9$$

15)

$R\cos(x-\alpha) = R\cos x\cos\alpha + R\sin x\sin\alpha$

So $R\cos\alpha = 2$ and $R\sin\alpha = \sqrt{5} \therefore R = 3$ and $\tan\alpha = \frac{\sqrt{5}}{2}$

$\therefore y = \frac{8}{3\cos(x-\alpha)}$

$$\pi\int_0^{\frac{\pi}{4}} y^2\,\mathrm{d}x = \pi\int_0^{\frac{\pi}{4}}\left(\frac{8}{2\cos x+\sqrt{5}\sin x}\right)^2\mathrm{d}x = \pi\int_0^{\frac{\pi}{4}}\left(\frac{8}{3\cos(x-\alpha)}\right)^2\mathrm{d}x$$

$$= \frac{64}{9}\pi\int_0^{\frac{\pi}{4}}\left(\frac{1}{\cos^2(x-\alpha)}\right)\mathrm{d}x$$

$$= \frac{64}{9}\pi\int_0^{\frac{\pi}{4}}\sec^2(x-\alpha)\,\mathrm{d}x$$

$$= \frac{64}{9}\pi[\tan(x-\alpha)]_0^{\frac{\pi}{4}}$$

$$= \frac{64}{9}\pi\left[\left(\tan\left(\frac{\pi}{4}-\alpha\right)\right)-(\tan(-\alpha))\right]$$

Since $\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A\tan B}$ then:

$$\tan\left(\frac{\pi}{4}-\alpha\right) = \frac{\tan\frac{\pi}{4}-\frac{\sqrt{5}}{2}}{1+\frac{\sqrt{5}}{2}\tan\frac{\pi}{4}} = \frac{1-\frac{\sqrt{5}}{2}}{1+\frac{\sqrt{5}}{2}} = \frac{2-\sqrt{5}}{2+\sqrt{5}} = 4\sqrt{5}-9$$

$$\tan(0-\alpha) = -\tan\alpha = -\frac{\sqrt{5}}{2}$$

Which gives:

$$\frac{64}{9}\pi\left[\left(\tan\left(\frac{\pi}{4}-\alpha\right)\right)-(\tan(-\alpha))\right] = \frac{64}{9}\pi\left[4\sqrt{5}-9+\frac{\sqrt{5}}{2}\right] = \frac{64}{9}\pi\left[-9+\frac{9\sqrt{5}}{2}\right]$$

$= 32\pi(\sqrt{5}-2)$ units^3 as req.

16) a)

$\left(z-\frac{1}{z}\right)^6 = z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}$

$= \left(z^6 + \frac{1}{z^6}\right) - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20$

Since $z - \frac{1}{z} = 2\mathrm{i}\sin\theta$ and $z^n + z^{-n} = 2\cos n\theta$ then

$\left(z-\frac{1}{z}\right)^6 = 2\cos 6\theta - 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta - 20$

So $(2\mathrm{i}\sin\theta)^6 = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$

$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$

$\sin^6\theta = -\frac{2}{64}\cos 6\theta + \frac{12}{64}\cos 4\theta - \frac{30}{64}\cos 2\theta + \frac{20}{64}$

$\therefore \sin^6\theta = \frac{5}{16} - \frac{15}{32}\cos 2\theta + \frac{3}{16}\cos 4\theta - \frac{1}{32}\cos 6\theta$ as req.

16) b)

$H = 3$

16) c)

$\pi\int_0^4 y^2\,\mathrm{d}x = \pi\int_0^4 \left(3\sin^3\left(\frac{\pi x}{8}\right)\right)^2 \mathrm{d}x$

$= 9\pi\int_0^4 \left(\sin^6\left(\frac{\pi x}{8}\right)\right)\mathrm{d}x$

$= 9\pi\int_0^4 \left(\frac{5}{16} - \frac{15}{32}\cos\frac{\pi x}{4} + \frac{3}{16}\cos\frac{\pi x}{2} - \frac{1}{32}\cos\frac{3\pi x}{4}\right)\mathrm{d}x$

$= 9\pi\left[\frac{5}{16}x - \frac{15}{8\pi}\sin\frac{\pi x}{4} + \frac{3}{8\pi}\sin\frac{\pi x}{2} - \frac{1}{24\pi}\sin\frac{3\pi x}{4}\right]_0^4$

$= 9\pi\left[\left(\frac{5}{4} - 0 + 0 - 0\right) - (0 - 0 + 0 - 0)\right]$

$= \frac{45}{4}\pi\ \mathrm{cm}^3$

16) d)

The model assumes the ornament is a perfectly smooth solid of revolution with no wall thickness.

17)

$x = \frac{1}{2\sqrt{2}-x} \Rightarrow x(2\sqrt{2}-x) = 1 \Rightarrow 2\sqrt{2}x - x^2 = 1 \Rightarrow x^2 - 2\sqrt{2}x + 1 = 0$

Solving for x:

$(x-\sqrt{2})^2 - 2 + 1 = 0 \Rightarrow (x-\sqrt{2})^2 = 1 \Rightarrow x - \sqrt{2} = \pm 1$

$\therefore x = 1+\sqrt{2}$ or $x = -1+\sqrt{2}$

Since $y^2 = \left(\frac{1}{2\sqrt{2}-x}\right)^2 = (2\sqrt{2}-x)^{-2}$

$$\pi \int_{-1+\sqrt{2}}^{1+\sqrt{2}} y^2 \, \mathrm{d}x = \pi \int_{-1+\sqrt{2}}^{1+\sqrt{2}} \left[x^2 - (2\sqrt{2}-x)^{-2}\right] \mathrm{d}x$$

$$= \pi \left[\frac{1}{3}x^3 - \frac{1}{2\sqrt{2}-x}\right]_{-1+\sqrt{2}}^{1+\sqrt{2}}$$

$$= \pi \left[\left(\frac{1}{3}x^3 - \frac{1}{2\sqrt{2}-(1+\sqrt{2})}\right) - \left(\frac{1}{3}x^3 - \frac{1}{2\sqrt{2}-(-1+\sqrt{2})}\right)\right]$$

$$= \pi \left[\left(\frac{1}{3}(1+\sqrt{2})^3 - \frac{1}{\sqrt{2}-1}\right) - \left(\frac{1}{3}(-1+\sqrt{2})^3 - \frac{1}{\sqrt{2}+1}\right)\right]$$

$$= \pi \left[\left(\frac{1}{3}(7+5\sqrt{2}) - \frac{\sqrt{2}+1}{1}\right) - \left(\frac{1}{3}(-7+5\sqrt{2}) - \frac{\sqrt{2}-1}{1}\right)\right]$$

$$= \pi \left[\frac{7}{3} + \frac{5}{3}\sqrt{2} - \sqrt{2} - 1 + \frac{7}{3} - \frac{5}{3}\sqrt{2} + \sqrt{2} - 1\right]$$

$= \frac{8}{3}\pi$ units^3

18) a)

$r = \frac{3}{2-\cos\theta} \Rightarrow 2r - r\cos\theta = 3$

Since $r\cos\theta = x$

$2r - x = 3 \Rightarrow r = \frac{x+3}{2}$

Square both sides: $r^2 = \frac{(x+3)^2}{4}$

Replace r^2: $x^2 + y^2 = \frac{(x+3)^2}{4}$

$\therefore y^2 = \frac{(x+3)^2}{4} - x^2$ as req.

18) b)

Consider the point $(x, -y)$:

$(-y)^2 = \frac{(x+3)^2}{4} - x^2 \Rightarrow y^2 = \frac{(x+3)^2}{4} - x^2$

Therefore whenever a point (x, y) is on the curve, the point $(x, -y)$ is also on the curve. Hence the curve is symmetrical about the x-axis.

18) c)

$\frac{(x+3)^2}{4} - x^2 = 0 \Rightarrow x^2 + 6x + 9 = 4x^2 \Rightarrow 3x^2 - 6x - 9 = 0$ so $x = -1$ or $x = 3$

$$\pi \int_0^3 y^2 \, \mathrm{d}x = \pi \int_0^3 \left(\frac{(x+3)^2}{4} - x^2 \right) \mathrm{d}x$$

$$= \pi \int_0^3 \left(-\frac{3}{4}x^2 + \frac{3}{2}x + \frac{9}{4} \right) \mathrm{d}x$$

$$= \pi \left[-\frac{1}{4}x^3 + \frac{3}{4}x^2 + \frac{9}{4}x \right]_0^3$$

$$= \pi \left[\left(-\tfrac{1}{4} \times 27 + \tfrac{3}{4} \times 9 + \tfrac{9}{4} \times 3 \right) - (0) \right]$$

$$= \frac{27}{4}\pi \text{ units}^3$$

19) a)

$r = 3 - 2\sec\theta \Rightarrow r = 3 - \frac{2}{\cos\theta}$

$r\cos\theta = 3\cos\theta - 2$

$r^2\cos\theta = 3r\cos\theta - 2r$

Since $r\cos\theta = x$

$xr = 3x - 2r \Rightarrow xr + 2r = 3x \Rightarrow r(x+2) = 3x \Rightarrow r = \frac{3x}{x+2}$

Square both sides: $r^2 = \frac{9x^2}{(x+2)^2}$

Replace r^2: $x^2 + y^2 = \frac{9x^2}{(x+2)^2}$

$\therefore y^2 = \frac{9x^2}{(x+2)^2} - x^2$ as req.

19) b)

Consider the point $(x, -y)$:

$(-y)^2 = \frac{9x^2}{(x-2)^2} - x^2 \Rightarrow y^2 = \frac{9x^2}{(x-2)^2} - x^2$

Therefore whenever a point (x, y) is on the curve, the point $(x, -y)$ is also on the curve. Hence the curve is symmetrical about the x-axis.

19) c)

Simplify using algebraic division:

$y^2 = \frac{9x^2}{(x+2)^2} - x^2 = \frac{9x^2}{x^2+4x+4} - x^2 = \frac{9(x^2+4x+4)-36x-36}{x^2+4x+4} - x^2 = 9 + \frac{-36x-36}{(x+2)^2} - x^2$

Partial fractions on $\frac{-36x-36}{(x+2)^2}$

$\frac{-36x-36}{(x+2)^2} \equiv \frac{A}{x+2} + \frac{B}{(x+2)^2}$

$-36x - 36 = A(x+2) + B$

Let $x = -2$: $\quad 36 = B$

Let $x = 0$: $\quad -36 = 2A + B$

$\therefore A = -36$ and $B = 36$

$\frac{-36x-36}{(x+2)^2} = \frac{36}{(x+2)^2} - \frac{36}{x+2}$

$\therefore y^2 = 9 + \frac{36}{(x+2)^2} - \frac{36}{x+2} - x^2$

19) d)

$$\pi \int_0^1 y^2 \, \mathrm{d}x = \pi \int_0^1 \left(\frac{9x^2}{(x+2)^2} - x^2 \right) \mathrm{d}x$$

$$= \pi \int_0^1 \left(9 + \frac{36}{(x+2)^2} - \frac{36}{x+2} - x^2 \right) \mathrm{d}x$$

$$= \pi \left[9x - \frac{36}{x+2} - 36\ln|x+2| - \frac{1}{3}x^3 \right]_0^1$$

$$= \pi \left[\left(9 - 12 - 36\ln 3 - \frac{1}{3} \right) - (0 - 18 - 36\ln 2 - 0) \right]$$

$$= \pi \left[\frac{44}{3} + 36\ln 2 - 36\ln 3 \right]$$

$$= \pi \left(\frac{44}{3} + 36\ln \frac{2}{3} \right) \text{ units}^3$$

20) a)

$$\pi \int_0^{\frac{1}{2}} y^2 \, \mathrm{d}x = \pi \int_0^{\frac{1}{2}} \left((1-x^2)^{-\frac{1}{4}}\right)^2 \, \mathrm{d}x$$

$$= \pi \int_0^{\frac{1}{2}} \left(\frac{1}{\sqrt{1-x^2}}\right) \mathrm{d}x$$

$$= \pi [\arcsin x]_0^{\frac{1}{2}}$$

$$= \pi \left[\arcsin \frac{1}{2} - \arcsin 0\right]$$

$$= \frac{\pi^2}{6} \text{ units}^3$$

20) b)

$$\pi \int_0^4 y^2 \, \mathrm{d}x = \pi \int_0^4 \left(\frac{3}{\sqrt{x^2+9}}\right)^2 \, \mathrm{d}x$$

$$= \pi \int_0^4 \left(\frac{9}{x^2+9}\right) \mathrm{d}x$$

$$= 9\pi \int_0^4 \left(\frac{1}{x^2+9}\right) \mathrm{d}x$$

$$= 9\pi \left[\frac{1}{3} \arctan\left(\frac{x}{3}\right)\right]_0^4$$

$$= 9\pi \left(\frac{1}{3} \arctan\left(\frac{4}{3}\right) - \frac{1}{3} \arctan(0)\right)$$

$$= 3\pi \arctan\left(\frac{4}{3}\right) \text{ units}^3$$

20) c)

$$\pi\int_0^{\frac{1}{4}} y^2\,\mathrm{d}x = \pi\int_0^{\frac{1}{4}}\left((8-4x-4x^2)^{-\frac{1}{4}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^{\frac{1}{4}}\left(\frac{1}{\sqrt{8-4x-4x^2}}\right)\mathrm{d}x$$

$$= \pi\int_0^{\frac{1}{4}}\left(\frac{1}{\sqrt{9-4\left(x+\frac{1}{2}\right)^2}}\right)\mathrm{d}x$$

$$= \frac{\pi}{3}\int_0^{\frac{1}{4}}\left(\frac{1}{\sqrt{1-\left(\frac{2}{3}x+\frac{1}{3}\right)^2}}\right)\mathrm{d}x$$

$$= \frac{\pi}{3}\left[\frac{3}{2}\arcsin\left(\frac{2}{3}x+\frac{1}{3}\right)\right]_0^{\frac{1}{4}}$$

$$= \frac{\pi}{3}\left(\frac{3}{2}\arcsin\frac{1}{2}-\frac{3}{2}\arcsin\frac{1}{3}\right)$$

$$= \frac{\pi}{3}\left(\frac{\pi}{4}-\frac{3}{2}\arcsin\frac{1}{3}\right)$$

$$= \frac{\pi}{12}\left(\pi-6\arcsin\frac{1}{3}\right)\text{ units}^3$$

20) d)

$$\pi\int_0^{\sqrt{6}} y^2\,\mathrm{d}x = \pi\int_0^{\sqrt{6}}\left(\frac{1}{\sqrt{x^2+2}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^{\sqrt{6}}\left(\frac{1}{2+x^2}\right)\mathrm{d}x$$

$$= \pi\left[\frac{1}{\sqrt{2}}\arctan\left(\frac{x}{\sqrt{2}}\right)\right]_0^{\sqrt{6}}$$

$$= \pi\left[\frac{1}{\sqrt{2}}\arctan\left(\frac{\sqrt{6}}{\sqrt{2}}\right)-\frac{1}{\sqrt{2}}\arctan(0)\right]$$

$$= \pi\left(\frac{\pi}{3\sqrt{2}}-0\right)\text{ units}^3$$

$$= \frac{\pi^2}{3\sqrt{2}}\text{ units}^3$$

20) e)

$$\pi\int_0^1 y^2\,\mathrm{d}x = \pi\int_0^1 \left(\frac{1}{9+x^2}\right)^2\,\mathrm{d}x$$

$$= \frac{\pi}{81}\int_0^1 \left(\frac{1}{1+\left(\frac{x}{3}\right)^2}\right)^2\,\mathrm{d}x$$

Let $x = 3\tan\theta$ then $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 3\sec^2\theta$

When $x = 0$, $\theta = 0$ and $x = 1$, $\theta = \arctan\left(\frac{1}{3}\right)$

So that

$$\frac{\pi}{81}\int_0^1 \left(\frac{1}{1+\left(\frac{x}{3}\right)^2}\right)^2\,\mathrm{d}x = \frac{\pi}{81}\int_0^{\arctan\left(\frac{1}{3}\right)} \left(\frac{1}{1+\tan^2\theta}\right)^2 \times 3\sec^2\theta\,\mathrm{d}\theta$$

$$= \frac{\pi}{27}\int_0^{\arctan\left(\frac{1}{3}\right)} \frac{1}{\sec^2\theta}\,\mathrm{d}\theta$$

$$= \frac{\pi}{27}\int_0^{\arctan\left(\frac{1}{3}\right)} \cos^2\theta\,\mathrm{d}\theta$$

$$= \frac{\pi}{27}\int_0^{\arctan\left(\frac{1}{3}\right)} \left(\frac{1}{2}+\frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$$

$$= \frac{\pi}{27}\left[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right]_0^{\arctan\left(\frac{1}{3}\right)}$$

$$= \frac{\pi}{27}\left[\left(\frac{1}{2}\arctan\left(\frac{1}{3}\right) + \frac{1}{4}\sin\left(2\arctan\left(\frac{1}{3}\right)\right)\right) - (0+0)\right]$$

$$= \frac{\pi}{27}\left(\frac{1}{2}\arctan\left(\frac{1}{3}\right) + \frac{3}{20}\right)\text{ units}^3$$

$$= \frac{\pi}{54}\left(\arctan\left(\frac{1}{3}\right) + \frac{3}{10}\right)\text{ units}^3$$

21)

$$\pi\int_0^a y^2\,\mathrm{d}x = \pi\int_0^a \left(\frac{1}{\sqrt{4-x^2}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^a \left(\frac{1}{4-x^2}\right)\mathrm{d}x$$

$$\frac{1}{4-x^2} = \frac{A}{2-x} + \frac{B}{2+x}$$

$$1 = A(2+x) + B(2-x)$$

When $x = 2$: $\quad 1 = 4A$

When $x = -2$: $\quad 1 = 4B$

$$= \frac{1}{4}\pi\int_0^1 \left(\frac{1}{2-x} + \frac{1}{2+x}\right)\mathrm{d}x$$

$$= \tfrac{1}{4}\pi[-\ln|2-x| + \ln|2+x|]_0^1$$

$$= \tfrac{1}{4}\pi\left[\ln\left|\tfrac{2+x}{2-x}\right|\right]_0^1$$

$$= \tfrac{1}{4}\pi(\ln 3)\text{ units}^3$$

Alternatively:

$$\pi\int_0^1 y^2\,\mathrm{d}x = \pi\int_0^1 \left(\frac{1}{\sqrt{4-x^2}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^1 \left(\frac{1}{4-x^2}\right)\mathrm{d}x$$

$$= \pi\left[\tfrac{1}{2}\operatorname{artanh}\left(\tfrac{x}{2}\right)\right]_0^1$$

$$= \pi\left(\tfrac{1}{2}\operatorname{artanh}\left(\tfrac{1}{2}\right) - \tfrac{1}{2}\operatorname{artanh}(0)\right)$$

$$= \tfrac{1}{2}\pi \times \tfrac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)$$

$$= \tfrac{1}{4}\pi(\ln 3)\text{ units}^3$$

22)

$$\pi\int_0^2 y^2\,\mathrm{d}x = \pi\int_0^2 \left(\frac{1}{(x^2+3)^{\frac{1}{4}}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^2 \left(\frac{1}{\sqrt{x^2+3}}\right)\mathrm{d}x$$

$$= \pi\int_0^2 \left(\frac{1}{\sqrt{3}\sqrt{1+\left(\frac{x}{\sqrt{3}}\right)^2}}\right)\mathrm{d}x$$

$$= \pi\left[\operatorname{arsinh}\left(\frac{x}{\sqrt{3}}\right)\right]_0^2$$

$$= \pi\left[\operatorname{arsinh}\left(\frac{2}{\sqrt{3}}\right) - \operatorname{arsinh} 0\right]$$

$$= \pi\left(\ln\left(\frac{2}{\sqrt{3}} + \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2+1}\right)\right)$$

$$= \pi\left(\ln\left(\frac{2}{\sqrt{3}} + \sqrt{\frac{7}{3}}\right)\right)$$

$$= \pi\left(\ln\left(\frac{2+\sqrt{7}}{\sqrt{3}}\right)\right)\text{ units}^3$$

23) a)

Either model must go through: $(0, 2)$ and $(2, 0.4)$

$2 = 0.2 \times 0^2 + 0 + b$ so $b = 2$

$0.4 = 0.2 \times 2^2 + 2a + 2 \Rightarrow -2.4 = 2a$ so $a = -1.2$

$a = -1.2$ and $b = 2$ as req.

23) b)

Either model must go through: $(0, 2)$ and $(2, 0.4)$

$2 = \frac{p}{q}$ and $0.4 = \frac{p}{2^2+q}$

$2q = p$ and $1.6 + 0.4q = p$

So $2q = 1.6 + 0.4q \therefore q = 1$ and $p = 2$

23) c)

$$\pi\int_0^2 y^2\,\mathrm{d}x = \pi\int_0^2 (0.2x^2 - 1.2x + 2)^2\,\mathrm{d}x$$

$$= \pi\int_0^2 (4 - 4.8x + 2.24x^2 - 0.48x^3 + 0.04x^4)\,\mathrm{d}x$$

$$= \pi\left[4x - 2.4x^2 + \frac{56}{75}x^3 - 0.12x^4 + 0.008x^5\right]_0^2$$

$$= \pi[2.70933\ldots - 0]$$

$$= 8.51\ \mathrm{m}^3$$

23) d)

$$\pi\int_0^2 y^2\,\mathrm{d}x = \pi\int_0^2 \left(\frac{2}{1+x^2}\right)^2\,\mathrm{d}x$$

$$= 4\pi\int_0^2 \left(\frac{1}{1+x^2}\right)^2\,\mathrm{d}x$$

Let $x = \tan\theta$ then $\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta$

When $x = 0$, $\theta = 0$ and $x = 2$, $\theta = \arctan(2)$

So that

$$4\pi\int_0^2 \left(\frac{1}{1+x^2}\right)^2\,\mathrm{d}x = 4\pi\int_0^{\arctan(2)} \left(\frac{1}{1+\tan^2\theta}\right)^2 \times \sec^2\theta\,\mathrm{d}\theta$$

$$= 4\pi\int_0^{\arctan(2)} \frac{1}{\sec^2\theta}\,\mathrm{d}\theta$$

$$= 4\pi\int_0^{\arctan(2)} \cos^2\theta\,\mathrm{d}\theta$$

$$= 4\pi\int_0^{\arctan(2)} \left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$$

$$= 4\pi\left[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right]_0^{\arctan(2)}$$

$$= 4\pi\left[\left(\tfrac{1}{2}\arctan(2) + \tfrac{1}{4}\sin(2\arctan(2))\right) - (0+0)\right]$$

$$= 9.47\ \mathrm{m}^3$$

24)

$$\pi\int_1^2 y^2\,\mathrm{d}x = \pi\int_1^2 \operatorname{arcosh}(x)\;\mathrm{d}x$$

Use Integration by Parts:

$$u = \operatorname{arcosh}(x) \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}} \qquad v = x$$

$$\therefore \int \operatorname{arcosh}(x)\,\mathrm{d}x = x\operatorname{arcosh}(x) - \int \frac{x}{\sqrt{x^2-1}}\mathrm{d}x = x\operatorname{arcosh}(x) - \sqrt{x^2-1}(+c)$$

So that

$$\pi\int_1^2 \operatorname{arcosh}(x)\;\mathrm{d}x = \pi\left[x\operatorname{arcosh}(x) - \sqrt{x^2-1}\right]_1^2$$

$$= \pi\left[\left(2\operatorname{arcosh}(2) - \sqrt{3}\right) - (0-0)\right]$$

$$= \pi\left(2\ln\left(2+\sqrt{3}\right) - \sqrt{3}\right) \text{ units}^3 \text{ as req.}$$

25) a)

$$\pi\int_0^k y^2\,\mathrm{d}x = \pi\int_0^k \left(\frac{1}{(k-x^2)^{\frac{1}{4}}}\right)^2\,\mathrm{d}x$$

$$= \pi\int_0^k \left(\frac{1}{\sqrt{k-x^2}}\right)\mathrm{d}x$$

$$= \pi\left[\arcsin\left(\frac{x}{\sqrt{k}}\right)\right]_0^k$$

$$= \pi\left[\arcsin\left(\frac{k}{\sqrt{k}}\right) - \arcsin(0)\right]$$

$$= \pi\arcsin\left(\sqrt{k}\right) \text{ units}^3$$

25) b)

$$\pi \int_0^{\frac{1}{2}} y^2 \, \mathrm{d}x$$

$$= \pi \left[\arcsin\left(\frac{x}{\sqrt{k}}\right) \right]_0^{\frac{1}{2}}$$

$$= \pi \left[\arcsin\left(\frac{\frac{1}{2}}{\sqrt{k}}\right) - \arcsin(0) \right]$$

$$= \pi \arcsin\left(\frac{1}{2\sqrt{k}}\right)$$

$$= \pi \arcsin\left(\frac{\sqrt{k}}{2k}\right) \text{ units}^3$$

26)

When $y = 0$:

$0 = \sqrt{2}x + \sqrt{2} \Rightarrow x = -1$

$0 = -\frac{x-2}{\sqrt{x^2+2}} \Rightarrow x = 2$

Total volume is given by:

$$\pi \int_{-1}^{0} \left(\sqrt{2}x + \sqrt{2}\right)^2 \mathrm{d}x + \pi \int_{0}^{2} \left(-\frac{x-2}{\sqrt{x^2+2}}\right)^2 \mathrm{d}x$$

Each part:

$$\int_{-1}^{0} \left(\sqrt{2}x + \sqrt{2}\right)^2 \mathrm{d}x = \int_{-1}^{0} (2x^2 + 4x + 2)\,\mathrm{d}x$$

$$= \left[\frac{2}{3}x^3 + 2x^2 + 2x\right]_{-1}^{0} = (0) - \left(-\frac{2}{3} + 2 - 2\right) = \frac{2}{3}$$

$$\int_{0}^{2} \left(-\frac{x-2}{\sqrt{x^2+2}}\right)^2 \mathrm{d}x = \int_{0}^{2} \frac{(x-2)^2}{x^2+2}\mathrm{d}x$$

But since $\frac{(x-2)^2}{x^2+2} = \frac{x^2-4x+4}{x^2+2} = \frac{x^2+2-4x+2}{x^2+2} = 1 + \frac{-4x+2}{x^2+2}$

$$\int_{0}^{2} \frac{(x-2)^2}{x^2+2}\mathrm{d}x = \int_{0}^{2} \left(1 + \frac{-4x+2}{x^2+2}\right)\mathrm{d}x = \int_{0}^{2} \left(1 - \frac{4x}{x^2+2} + \frac{2}{x^2+2}\right)\mathrm{d}x$$

$$= \left[x - 2\ln|x^2+2| + \sqrt{2}\arctan\left(\frac{x}{\sqrt{2}}\right)\right]_{0}^{2}$$

$$= \left(2 - 2\ln 6 + \sqrt{2}\arctan\sqrt{2}\right) - (0 - 2\ln 2 + 0)$$

$$= 2 - \ln 36 + \ln 4 + \sqrt{2}\arctan\sqrt{2}$$

$$= 2 - \ln 9 + \sqrt{2}\arctan\sqrt{2}$$

So that

$$\pi \int_{-1}^{0} \left(\sqrt{2}x + \sqrt{2}\right)^2 \mathrm{d}x + \pi \int_{0}^{2} \left(-\frac{x-2}{\sqrt{x^2+2}}\right)^2 \mathrm{d}x$$

$$= \frac{2}{3}\pi + \left(2 - \ln 9 + \sqrt{2}\arctan\sqrt{2}\right)\pi$$

$$= \pi\left(\frac{8}{3} - \ln 9 + \sqrt{2}\arctan\sqrt{2}\right)$$

$= 5.72$ units^3 (3 SF)

27) a)

$$\pi\int_0^{\frac{7}{2}} x^2\,\mathrm{d}y = \pi\int_0^{\frac{7}{2}} (2y+1)^{\frac{2}{3}}\,\mathrm{d}y$$

$$= \pi\left[\tfrac{3}{10}(2y+1)^{\frac{5}{3}}\right]_0^{\frac{7}{2}} = \pi\left[\left(\tfrac{3}{10}\times 8^{\frac{5}{3}}\right) - \left(\tfrac{3}{10}\right)\right] = \tfrac{93}{10}\pi \text{ units}^3$$

27) b)

$$\pi\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} x^2\,\mathrm{d}y = \pi\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \operatorname{cosec}^2 y\,\mathrm{d}y$$

$$= \pi[-\cot y]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \pi\left[-\cot\tfrac{\pi}{3} + \cot\tfrac{\pi}{6}\right] = \pi\left(\sqrt{3} - \tfrac{\sqrt{3}}{3}\right) = \tfrac{2\sqrt{3}}{3}\pi \text{ units}^3$$

27) c)

$$\pi\int_0^{\frac{\pi}{2}} x^2\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{2}} \frac{2\sin y}{1+\cos y}\,\mathrm{d}y$$

$$= \pi[-2\ln(1+\cos y)]_0^{\frac{\pi}{2}} = \pi[0 - (-2\ln(2))] = (\ln 4)\pi \text{ units}^3$$

27) d)

$$\pi\int_0^{\frac{\pi}{4}} x^2\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{4}} \left(\sqrt{y}\sec y\right)^2\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{4}} y\sec^2 y\,\mathrm{d}y$$

Use Integration by Parts:

$$u = y \qquad \frac{\mathrm{d}v}{\mathrm{d}y} = \sec^2 y$$

$$\frac{\mathrm{d}u}{\mathrm{d}y} = 1 \qquad v = \tan y$$

$$\therefore \int y\sec^2 y\,\mathrm{d}y = y\tan y - \int \tan y\,\mathrm{d}y$$

$$= y\tan y + \ln|\cos y| + c$$

So that:

$$\pi\int_0^{\frac{\pi}{4}} y\sec^2 y\,\mathrm{d}y = \pi[y\tan y + \ln|\cos y|]_0^{\frac{\pi}{4}} = \pi\left[\left(\tfrac{\pi}{4} + \ln\left|\tfrac{1}{\sqrt{2}}\right|\right) - 0\right] = \pi\left[\left(\tfrac{\pi}{4} - \ln|\sqrt{2}|\right) - 0\right]$$

$$= \pi\left(\tfrac{\pi}{4} - \tfrac{1}{2}\ln 2\right) \text{ units}^3$$

$$= \tfrac{\pi}{4}(\pi - \ln 4) \text{ units}^3$$

27) e)

$$\pi\int_0^{\frac{\pi}{4}} x^2\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{4}} (\sec y - \tan y)^2\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{4}} (\sec^2 y - 2\sec y\tan y + \tan^2 y)\,\mathrm{d}y$$

$$= \pi\int_0^{\frac{\pi}{4}} (\sec^2 y - 2\sec y\tan y + \sec^2 y - 1)\,\mathrm{d}y = \pi\int_0^{\frac{\pi}{4}} (2\sec^2 y - 2\sec y\tan y - 1)\,\mathrm{d}y$$

$$= \pi[2\tan y - 2\sec y - y]_0^{\frac{\pi}{4}}$$

$$= \pi\left[\left(2 - \frac{4}{\sqrt{2}} - \frac{\pi}{4}\right) - (0 - 2 - 0)\right] = \left(4 - 2\sqrt{2} - \frac{\pi}{4}\right)\pi \text{ units}^3$$

28) a)

Since $x^2 = \frac{5}{9} - \frac{2}{9}y^2 + \frac{1}{9}\cos\left(\frac{\pi y}{3}\right)$

$$\pi\int_{-1.570}^{1.570} x^2\,\mathrm{d}y = \pi\int_{-1.570}^{1.570} \left(\frac{5}{9} - \frac{2}{9}y^2 + \frac{1}{9}\cos\left(\frac{\pi y}{3}\right)\right)\mathrm{d}y$$

$$= \pi\left[\frac{5}{9}y - \frac{2}{27}y^3 + \frac{1}{3\pi}\sin\left(\frac{\pi y}{3}\right)\right]_{-1.570}^{1.570} = \pi[0.691382\ldots - (-0.691382\ldots)] = 4.34 \text{ cm}^3$$

28) b)

$50 \times 4.34 \text{ cm}^3 = 217 \text{ cm}^3$

So she does not have enough Resin.

29)

$$y = \ln x \Rightarrow \mathrm{e}^y = x \Rightarrow \mathrm{e}^{2y} = x^2$$

$$\pi\int_0^{\ln 4} x^2\,\mathrm{d}y = \pi\int_0^{\ln 4} \mathrm{e}^{2y}\,\mathrm{d}y$$

$$= \pi\left[\frac{1}{2}\mathrm{e}^{2y}\right]_0^{\ln 4} = \pi\left[\frac{1}{2}\mathrm{e}^{2\ln 4} - \frac{1}{2}\mathrm{e}^0\right] = \pi\left(8 - \frac{1}{2}\right) = \frac{15}{2}\pi \text{ units}^3$$

30)

Since $x^2 = 1 - \sin\left(\frac{y}{h}\right)$

$$\pi \int_0^{\frac{\pi h}{2}} x^2 \, \mathrm{d}y = \pi \int_0^{\frac{\pi h}{2}} \left(1 - \sin\left(\frac{y}{h}\right)\right) \mathrm{d}y$$

$$= \pi \left[y + h\cos\left(\frac{y}{h}\right)\right]_0^{\frac{\pi h}{2}} = \pi \left[\left(\frac{\pi h}{2} - 0\right) + (0 - h)\right]$$

$$= \frac{\pi^2 h}{2} - \pi h$$

So $2\pi^2 - 4\pi = \frac{\pi^2 h}{2} - \pi h \therefore h = 4$ metres.

31) a)

$\sqrt{6} - \sqrt{2} = 4\sin\left(\frac{1}{2}y + \frac{\pi}{6}\right) \Rightarrow \frac{1}{2}y + \frac{\pi}{6} = \arcsin\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) \Rightarrow y = 2\arcsin\left(\frac{\sqrt{6}-\sqrt{2}}{4}\right) - \frac{\pi}{3} = -\frac{\pi}{6}$

$\frac{1}{2} \times -\frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{12}$

$\pi - \frac{\pi}{12} = \frac{11\pi}{12}$

$\frac{1}{2}y + \frac{\pi}{6} = \frac{11\pi}{12} \Rightarrow y = \frac{3\pi}{2}$

$\therefore k = \frac{3\pi}{2}$

31) b)

$x = 4\sin\left(\frac{1}{2} \times 0 + \frac{\pi}{6}\right) = 2$

So the diameter of the base is 4 cm.

31) c)

Volume of cylinder: $\pi\left(\sqrt{6}-\sqrt{2}\right)^2\left(6-\frac{3\pi}{2}\right)=4.33557797\ldots\ \text{cm}^3$

$$\pi\int_0^{\frac{3\pi}{2}}\left(4\sin\left(\frac{1}{2}y+\frac{\pi}{6}\right)\right)^2\mathrm{d}y=16\,\pi\int_0^{\frac{3\pi}{2}}\sin^2\left(\frac{1}{2}y+\frac{\pi}{6}\right)\mathrm{d}y$$

Since $\cos 2A=1-2\sin^2 A$ then $\sin^2 A=\frac{1}{2}-\frac{1}{2}\cos 2A$ so that

$$16\,\pi\int_0^{\frac{3\pi}{2}}\sin^2\left(\frac{1}{2}y+\frac{\pi}{6}\right)\mathrm{d}y=16\,\pi\int_0^{\frac{3\pi}{2}}\left(\frac{1}{2}-\frac{1}{2}\cos\left(y+\frac{\pi}{3}\right)\right)\mathrm{d}y$$

$$=16\,\pi\left[\frac{1}{2}y-\frac{1}{2}\sin\left(y+\frac{\pi}{3}\right)\right]_0^{\frac{3\pi}{2}}$$

$$=16\,\pi\left[\left(\frac{1}{2}\times\frac{3\pi}{2}-\frac{1}{2}\sin\left(\frac{3\pi}{2}+\frac{\pi}{3}\right)\right)-\left(0-\frac{1}{2}\sin\left(\frac{\pi}{3}\right)\right)\right]$$

$=152.757215\ldots\ \text{cm}^3$

So the total volume is $157\ \text{cm}^3$ (3 SF)

31) d)

The model ignores the thickness of the bottle walls.

32) a)

$x=k+2\sin\left(\frac{\pi y}{12}\right)$ at $(6,0)$ gives:

$6=k+2\sin(0)\ \therefore\ k=6$

32) b)

$x=6+2\sin\left(\frac{\pi y}{12}\right)$ when $x=8$

$\therefore\ 8=6+2\sin\left(\frac{\pi y}{12}\right)\Rightarrow\sin\left(\frac{\pi y}{12}\right)=1\ \therefore\ y=6$

So the height of the bowl is 6 cm.

32) c)

$$\pi\int_0^h x^2\,\mathrm{d}y = \pi\int_0^h \left(6+2\sin\left(\frac{\pi y}{12}\right)\right)^2\mathrm{d}y = \pi\int_0^h \left(36+24\sin\left(\frac{\pi y}{12}\right)+4\sin^2\left(\frac{\pi y}{12}\right)\right)\mathrm{d}y$$

Since $\cos 2A = 1-2\sin^2 A$ then $\sin^2 A = \frac{1}{2}-\frac{1}{2}\cos 2A$ so that

$$\pi\int_0^h \left(38+24\sin\left(\frac{\pi y}{12}\right)-2\cos\left(\frac{\pi y}{6}\right)\right)\mathrm{d}y$$

$$= \pi\left[38y-\frac{288}{\pi}\cos\left(\frac{\pi y}{12}\right)-\frac{12}{\pi}\sin\left(\frac{\pi y}{6}\right)\right]_0^h$$

$$= \pi\left[38h-\frac{288}{\pi}\cos\left(\frac{\pi h}{12}\right)-\frac{12}{\pi}\sin\left(\frac{\pi h}{6}\right)-\left(0-\frac{288}{\pi}\right)\right]$$

$$= 38\pi h-288\cos\left(\frac{\pi h}{12}\right)-12\sin\left(\frac{\pi h}{6}\right)+288 \text{ cm}^3$$

32) d)

$\frac{\mathrm{d}V}{\mathrm{d}t} = 500 \text{ cm}^3 \text{ min}^{-1}$

$$\frac{\mathrm{d}V}{\mathrm{d}h} = \pi\left(6+2\sin\left(\frac{\pi h}{12}\right)\right)^2$$

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}h}\times\frac{\mathrm{d}h}{\mathrm{d}t}$$

$$\therefore 500 = \pi\left(6+2\sin\left(\frac{\pi}{3}\right)\right)^2\times\frac{\mathrm{d}h}{\mathrm{d}t}$$

So that $\left.\frac{\mathrm{d}h}{\mathrm{d}t}\right|_{h=4} = \frac{500}{\pi\left(6+2\sin\left(\frac{\pi}{3}\right)\right)^2} = 2.66 \text{ cm min}^{-1}$

33) a)

$20x^2 + \ln(y+1) = 0.1$ at $x = 0 \Rightarrow \ln(y+1) = 0.1 \Rightarrow y = \mathrm{e}^{0.1} - 1$

Since $x^2 = \frac{1}{200} - \frac{1}{20}\ln(y+1)$

$\pi \int_0^{\mathrm{e}^{0.1}-1} x^2 \,\mathrm{d}y = \pi \int_0^{\mathrm{e}^{0.1}-1} \left(\frac{1}{200} - \frac{1}{20}\ln(y+1)\right) \mathrm{d}y$

Use integration by Parts:

$u = \ln(y+1)$ $\qquad$ $\frac{\mathrm{d}v}{\mathrm{d}y} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{1}{y+1}$ $\qquad$ $v = y$

$\therefore \int \ln(y+1)\,\mathrm{d}y = y\ln(y+1) - \int \frac{y}{y+1}\mathrm{d}y$

$= y\ln(y+1) - \int \frac{y+1-1}{y+1}\mathrm{d}y$

$= y\ln(y+1) - \int \left(1 - \frac{1}{y+1}\right)\mathrm{d}y$

$= y\ln(y+1) - y + \ln(y+1)\,(+c)$

$= (y+1)\ln(y+1) - y\,(+c)$

So that:

$\pi \int_0^{\mathrm{e}^{0.1}-1} \left(\frac{1}{200} - \frac{1}{20}\ln(y+1)\right) \mathrm{d}y$

$= \pi \left[\frac{1}{200}y - \frac{1}{20}(y+1)\ln(y+1) + \frac{1}{20}y\right]_0^{\mathrm{e}^{0.1}-1}$

$= \pi \left[\frac{11}{200}y - \frac{1}{20}(y+1)\ln(y+1)\right]_0^{\mathrm{e}^{0.1}-1}$

$= \pi \left[\left(\frac{11}{200}(\mathrm{e}^{0.1}-1) - \frac{1}{200}(\mathrm{e}^{0.1})\right) - (0)\right] = \left(\frac{1}{20}\mathrm{e}^{0.1} - \frac{11}{200}\right)\pi \text{ m}^3$

$= 0.000\,812 \text{ m}^3$

33) b)

$0.000\,812 \text{ m}^3 = 812 \text{ cm}^3$

mass = density × volume

$= 4.5 \times 812 = 3654 \text{ g} = 3.65 \text{ kg}$

33) c)

A mass of about 3.6 kg is quite heavy for a paperweight, suggesting the dimensions or density may be unrealistic for everyday use.

34)

$$\pi\int_0^{\sinh 1} x^2\,\mathrm{d}y = \pi\int_0^{\sinh 1} 2\,\mathrm{arsinh}\,y\,\mathrm{d}y$$

Use integration by Parts:

$$u = \mathrm{arsinh}\,y \qquad \frac{\mathrm{d}v}{\mathrm{d}y} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{1}{\sqrt{y^2+1}} \qquad v = y$$

$$\therefore \int \mathrm{arsinh}\,y\,\mathrm{d}y = y\,\mathrm{arsinh}\,y - \int \frac{y}{\sqrt{y^2+1}}\mathrm{d}y = y\,\mathrm{arsinh}\,y - \sqrt{y^2+1}\;(+c)$$

So that

$$\pi\int_0^{\sinh 1} 2\,\mathrm{arsinh}\,y\,\mathrm{d}y = 2\pi\left[y\,\mathrm{arsinh}\,y - \sqrt{y^2+1}\right]_0^{\sinh 1}$$

$$= 2\pi\left[\left(\sinh 1 - \sqrt{(\sinh 1)^2+1}\right) - (0-1)\right]$$

$$= 2\pi\left(1 + \sinh 1 - \sqrt{(\cosh 1)^2}\right)$$

$$= 2\pi\left(1 + \frac{1}{2}(\mathrm{e}^1 - \mathrm{e}^{-1}) - \frac{1}{2}(\mathrm{e}^1 + \mathrm{e}^{-1})\right)$$

$$= 2\pi\left(1 + \frac{1}{2}\mathrm{e}^1 - \frac{1}{2}\mathrm{e}^{-1} - \frac{1}{2}\mathrm{e}^1 - \frac{1}{2}\mathrm{e}^{-1}\right)$$

$$= 2\pi\left(1 - \frac{1}{\mathrm{e}}\right) \text{ as req.}$$

35)

For curve C_1: $x^2 = \frac{k^2}{k-10}y - \frac{10k^2}{k-10}$ for $k \leq y \leq 10$

For curve C_2: $x^2 = k^2\left(\frac{y}{k}\left(2-\frac{y}{k}\right)\right)^{\frac{1}{2}} = \frac{k^2}{\sqrt{k}}\sqrt{y\left(2-\frac{y}{k}\right)} = k^{\frac{3}{2}}\sqrt{2y-\frac{y^2}{k}} = k^{\frac{3}{2}}\sqrt{k-\frac{1}{k}(y-k)^2}$

$= k\sqrt{k^2-(y-k)^2}$

Consider each separately:

$\pi\int_k^{10}\left(\frac{k^2}{k-10}y - \frac{10k^2}{k-10}\right)\mathrm{d}y = \pi\left[\frac{k^2}{2k-20}y^2 - \frac{10k^2}{k-10}y\right]_k^{10}$

$= \pi\left[\left(\frac{100k^2}{2k-20} - \frac{100k^2}{k-10}\right) - \left(\frac{k^4}{2k-20} - \frac{10k^3}{k-10}\right)\right] = \pi\left[-\frac{50k^2}{k-10} - \frac{k^4-20k^3}{2k-20}\right] = \pi\left[\frac{-100k^2-k^4+20k^3}{2k-20}\right]$

$= -\frac{1}{2}k^2\pi\left[\frac{k^2-20k+100}{k-10}\right] = -\frac{1}{2}k^2\pi\left[\frac{(k-10)^2}{k-10}\right] = -\frac{1}{2}k^2(k-10)\pi$

And:

$\pi\int_0^k\left(k\sqrt{k^2-(y-k)^2}\right)\mathrm{d}y = \pi k\int_0^k\sqrt{k^2-(y-k)^2}\,\mathrm{d}y$

Let $y-k = k\sin\theta$ so $\frac{\mathrm{d}y}{\mathrm{d}\theta} = k\cos\theta \Rightarrow \frac{1}{k}\mathrm{d}y = \cos\theta\,\mathrm{d}\theta$

When $y = 0$, $\theta = -\frac{\pi}{2}$ and when $y = k$, $\theta = 0$

So

$\pi k\int_0^k\sqrt{k^2-(y-k)^2}\,\mathrm{d}y = \pi k\int_{\frac{3\pi}{2}}^0\sqrt{k^2-k^2\sin^2\theta}\,k\cos\theta\,\mathrm{d}\theta$

$= \pi k^3\int_{-\frac{\pi}{2}}^0\sqrt{1-\sin^2\theta}\cos\theta\,\mathrm{d}\theta$

$= \pi k^3\int_{-\frac{\pi}{2}}^0\cos^2\theta\,\mathrm{d}\theta = \pi k\int_{-\frac{\pi}{2}}^0\left(\frac{1}{2}+\frac{1}{2}\cos 2\theta\right)\mathrm{d}\theta$

$= k^3\pi\left[\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta\right]_{-\frac{\pi}{2}}^0$

$= k^3\pi\left[(0+0) - \left(-\frac{\pi}{4}\right)\right]$

$= \frac{1}{4}k^3\pi^2$

So the total volume is:

$-\frac{1}{2}k^2(k-10)\pi + \frac{1}{4}k^3\pi^2$

$= \frac{1}{4}k^2\pi\big(k\pi - 2(k-10)\big)$ units^3 as req.

36)

$y^2 = \frac{1}{x^2} - 1 \Rightarrow y^2 + 1 = \frac{1}{x^2} \Rightarrow x^2 = \frac{1}{1+y^2}$

$\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 \frac{1}{1+y^2} \mathrm{d}y = \pi[\arctan(y)]_0^1 = \pi[(\arctan(1)) - 0] = \frac{\pi^2}{4}$ units^3

37)

$$\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 \frac{1}{\sqrt{k - y^2}} \mathrm{d}y$$

$$= \pi \left[\arcsin\left(\frac{y}{\sqrt{k}}\right)\right]_0^1 = \pi \left[\left(\arcsin\left(\frac{1}{\sqrt{k}}\right)\right) - 0\right] = \pi \arcsin\left(\frac{1}{\sqrt{k}}\right) \text{ units}^3$$

38)

$x = \frac{1}{(k+y^2)^{\frac{1}{4}}} \Rightarrow x^2 = \frac{1}{\sqrt{k+y^2}}$

$$\pi \int_0^1 x^2 \, \mathrm{d}y = \pi \int_0^1 \frac{1}{\sqrt{k + y^2}} \mathrm{d}y$$

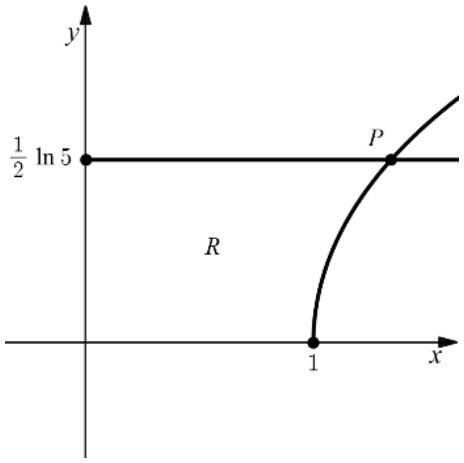
$$= \pi \left[\operatorname{arsinh}\left(\frac{y}{\sqrt{k}}\right)\right]_0^1 = \pi \left[\left(\operatorname{arsinh}\left(\frac{1}{\sqrt{k}}\right)\right) - 0\right] = \pi \operatorname{arsinh}\left(\frac{1}{\sqrt{k}}\right)$$

So since $\pi \ln\left(\frac{1+\sqrt{10}}{3}\right) = \pi \operatorname{arsinh}\left(\frac{1}{\sqrt{k}}\right) \Rightarrow \ln\left(\frac{1+\sqrt{10}}{3}\right) = \operatorname{arsinh}\left(\frac{1}{\sqrt{k}}\right) \Rightarrow$

$\sinh\left(\ln\left(\frac{1+\sqrt{10}}{3}\right)\right) = \frac{1}{\sqrt{k}} \Rightarrow \frac{1}{2}\left(\mathrm{e}^{\ln\left(\frac{1+\sqrt{10}}{3}\right)} - \mathrm{e}^{-\ln\left(\frac{1+\sqrt{10}}{3}\right)}\right) = \frac{1}{\sqrt{k}}$

$\Rightarrow \sqrt{k} = \frac{2}{\frac{1+\sqrt{10}}{3} - \frac{3}{1+\sqrt{10}}} \Rightarrow \sqrt{k} = 3 \therefore k = 9$

39) a)



39) b)

$\frac{1}{2}\ln 5 = \operatorname{arcosh} x$

$\Rightarrow x = \cosh\left(\frac{1}{2}\ln 5\right) = \frac{1}{2}\left(e^{\frac{1}{2}\ln 5} + e^{-\frac{1}{2}\ln 5}\right) = \frac{1}{2}\left(e^{\ln\sqrt{5}} + e^{\ln\frac{1}{\sqrt{5}}}\right) = \frac{1}{2}\left(\sqrt{5} + \frac{1}{\sqrt{5}}\right) = \frac{3\sqrt{5}}{5}$

$\therefore P\left(\frac{3\sqrt{5}}{5}, \frac{1}{2}\ln 5\right)$

39) c)

$y = \operatorname{arcosh} x \Rightarrow x^2 = \cosh^2 y$

$\pi\int_0^{\frac{1}{2}\ln 5} x^2\,\mathrm{d}y = \pi\int_0^{\frac{1}{2}\ln 5} \cosh^2 y\,\mathrm{d}y$

Since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$ so that

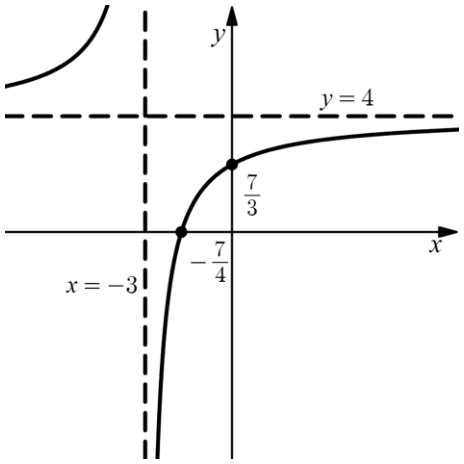
$\pi\int_0^{\frac{1}{2}\ln 5} \cosh^2 y\,\mathrm{d}y = \pi\int_0^{\frac{1}{2}\ln 5} \left(\frac{1}{2} + \frac{1}{2}\cosh 2y\right)\mathrm{d}y$

$= \pi\left[\frac{1}{2}y + \frac{1}{4}\sinh 2y\right]_0^{\frac{1}{2}\ln 5} = \pi\left[\left(\frac{1}{4}\ln 5 + \frac{1}{4}\sinh(\ln 5)\right) - (0+0)\right] = \pi\left(\frac{1}{4}\ln 5 + \frac{3}{5}\right)$ units^3.

40) a)

$a = 4$ and $b = 3$

40) b)



40) c)

$y = \frac{4x+7}{x+3} = \frac{4(x+3)-5}{x+3} = 4 - \frac{5}{x+3}$

$\pi \int_{-\frac{7}{4}}^{0} y^2 \, dx = \pi \int_{-\frac{7}{4}}^{0} \left(4 - \frac{5}{x+3}\right)^2 dx = \pi \int_{-\frac{7}{4}}^{0} \left(16 - \frac{40}{x+3} + \frac{25}{(x+3)^2}\right) dx$

$= \pi \left[16x - 40\ln|x+3| - \frac{25}{x+3}\right]_{-\frac{7}{4}}^{0} = \pi \left[\left(0 - 40\ln 3 - \frac{25}{3}\right) - \left(-28 - 40\ln\frac{5}{4} - 20\right)\right]$

$= \pi \left(\frac{119}{3} - 40\ln\frac{12}{5}\right)$ units^3.

40) d)

$y = \frac{4x+7}{x+3} \Rightarrow y(x+3) = 4x+7 \Rightarrow xy - 4x = 7 - 3y \Rightarrow x = \frac{7-3y}{y-4}$

$x = \frac{7-3y}{y-4} = \frac{-3(y-4)-5}{y-4} = -3 - \frac{5}{y-4}$

$\pi \int_{0}^{\frac{7}{3}} x^2 \, dy = \pi \int_{0}^{\frac{7}{3}} \left(-3 - \frac{5}{y-4}\right)^2 dy = \pi \int_{0}^{\frac{7}{3}} \left(9 + \frac{30}{y-4} + \frac{25}{(y-4)^2}\right) dy$

Note: $[\ln|y-4|]_a^b = \ln|b-4| - \ln|a-4| = \ln\left(\frac{|b-4|}{|a-4|}\right)$ so that $\ln\left(\frac{b-4}{a-4}\right) = \ln\left(\frac{4-b}{4-a}\right)$

$= \pi \left[9y + 30\ln|y-4| - \frac{25}{y-4}\right]_0^{\frac{7}{3}} = \pi \left[\left(21 + 30\ln\frac{5}{3} + 15\right) - \left(0 + 30\ln 4 + \frac{25}{4}\right)\right]$

$= \pi \left(\frac{119}{4} - 30\ln\frac{12}{5}\right)$ units^3.

41) a)

$\pi \int_{0}^{1} y^2 \, dx = \pi \int_{0}^{1} \left(\sin\left(\frac{\pi x}{2}\right)\right)^2 dx$

Since $\cos 2A = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$

$\pi \int_{0}^{1} \left(\sin\left(\frac{\pi x}{2}\right)\right)^2 dx = \pi \int_{0}^{1} \left(\frac{1}{2} - \frac{1}{2}\cos(\pi x)\right) dx$

$= \pi \left[\frac{1}{2}x - \frac{1}{2\pi}\sin(\pi x)\right]_0^1 = \pi \left[\left(\frac{1}{2} - 0\right) - (0 - 0)\right]$

$= \frac{1}{2}\pi$ units^3 as req.

41) b)

$\arcsin y = \frac{\pi}{2}x \Rightarrow x = \frac{2}{\pi}\arcsin y$

$$\pi\int_0^1 x^2\,\mathrm{d}y = \pi\int_0^1 \left(\frac{2}{\pi}\arcsin y\right)^2 \mathrm{d}y = \frac{4}{\pi}\int_0^1 \arcsin^2 y\,\mathrm{d}y$$

Consider $\int \arcsin^2 y\,\mathrm{d}y$ only:

Note that:

$$y = \arcsin^2 x \Rightarrow y = (\arcsin x)^2 \Rightarrow \sqrt{y} = \arcsin x \Rightarrow \sin\sqrt{y} = x$$

$$\therefore \frac{1}{2\sqrt{y}}\cos\sqrt{y}\frac{\mathrm{d}y}{\mathrm{d}x} = 1 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\frac{1}{2\sqrt{y}}\cos\sqrt{y}} = \frac{2\sqrt{y}}{\cos\sqrt{y}} = \frac{2\arcsin x}{\sqrt{1-x^2}}$$

Use integration by Parts:

$$u = \arcsin^2 y \qquad \frac{\mathrm{d}v}{\mathrm{d}y} = 1$$

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\arcsin y}{\sqrt{1-y^2}} \qquad v = y$$

$$\therefore \int \arcsin^2 y\,\mathrm{d}y = y\arcsin^2 y - \int \frac{2y\arcsin y}{\sqrt{1-y^2}}\mathrm{d}y$$

Consider $\int \frac{2y\arcsin y}{\sqrt{1-y^2}}\mathrm{d}y$ only:

Let $y = \sin t \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t$

$t = \arcsin y$

$$\int \frac{2y\arcsin y}{\sqrt{1-y^2}}\mathrm{d}y = \int \frac{2t\sin t}{\sqrt{1-\sin^2 t}}\cos t\,\mathrm{d}t = \int \frac{2t\sin t}{\sqrt{\cos^2 t}}\cos t\,\mathrm{d}t = \int 2t\sin t\,\mathrm{d}t$$

Use integration by Parts:

$$u = 2t \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \sin t$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 2 \qquad v = -\cos t$$

$$\int 2t\sin t\,\mathrm{d}t = -2t\cos t + 2\int \cos t\,\mathrm{d}t = -2t\cos t + 2\sin t\ (+c)$$

And so:

$$\int \arcsin^2 y\,\mathrm{d}y = y\arcsin^2 y - \int \frac{2y\arcsin y}{\sqrt{1-y^2}}\mathrm{d}y$$

$$= y\arcsin^2 y + 2\arcsin y\cos(\arcsin y) - 2y\ (+c)$$

$$\frac{4}{\pi}\int_0^1 \arcsin^2 y\,\mathrm{d}y = \frac{4}{\pi}[y\arcsin^2 y + 2\arcsin y\cos(\arcsin y) - 2y]_0^1$$

$$= \frac{4}{\pi}\left[\left(\frac{\pi^2}{4} + 0 - 2\right) - (0 + 0 - 0)\right]$$

$$= \left(\pi - \frac{8}{\pi}\right) \text{ units}^3 \text{ as req.}$$